

PROJECT 1

Enterprise HQ Network

Design, Implementation and Technical Validation

Cisco Packet Tracer | Enterprise Network Infrastructure | Portfolio Project

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Project Type	Enterprise HQ Network Infrastructure
Platform	Cisco Packet Tracer
Scenario	TechSolutions - approximately 75 employees
Report Version	1.1 - Technical Review Edition
Date	2026

A practical enterprise networking project covering segmentation, Layer 3 switching, centralized services, security controls, monitoring, routing and PAT.

Executive Summary

This project documents the design and implementation of a secure enterprise headquarters network for the fictional company TechSolutions using Cisco Packet Tracer. The lab represents an organization of approximately seventy-five employees while using twelve client PCs as representative endpoints to keep the simulation manageable.

The network uses a hierarchical small-enterprise architecture with a Layer 3 core switch for inter-VLAN routing, a Layer 2 access switch for endpoint connectivity, and an edge router for the outside boundary and Port Address Translation (PAT). Five business/service VLANs separate HR, Finance, IT, Guest and Server traffic, while VLAN 999 is used as a parking VLAN for unused access ports.

Centralized infrastructure services include DHCP, DNS, an internal Web service, NTP and Syslog. DHCP requests are relayed from user VLANs to the centralized DHCP server through SVI helper addresses. Security controls include SSH-only remote management, extended ACLs, sticky-MAC port security, shutdown violation mode, disabled unused ports and the parking VLAN.

Validation was performed using live Packet Tracer outputs and end-to-end tests. The final report intentionally distinguishes between features fully validated in the lab and the limitations of the Packet Tracer Cloud used as the outside-network boundary. The project serves as Project 1 of a three-stage enterprise portfolio that will later introduce a branch office, VLSM, OSPF, redundancy and advanced Layer 2 security.

Portfolio positioning This is a hands-on implementation report, not a theoretical design only. Each major feature is supported by configuration or validation evidence captured from the working Packet Tracer project.

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Reading guide The report follows the network from business requirements to architecture, implementation, security, validation and lessons learned. Figures are captured from the final lab or produced as clearly labeled conceptual diagrams.

1. Project Overview

1.1 Company Background

TechSolutions is modeled as a medium-sized technology company with approximately seventy-five employees distributed across multiple departments. The headquarters requires secure segmentation, centralized services, controlled interdepartmental access, reliable internal routing and a design that can be expanded later without rebuilding the entire architecture.

Project 1 intentionally uses a single edge router, one Layer 3 core switch and one access switch. Device-level redundancy is not implemented at this stage; it is reserved for the later enterprise-expansion project.

1.2 Project Objectives

- Design a structured enterprise headquarters network.
- Segment departments using VLANs and a dedicated parking VLAN.
- Implement Layer 3 switching and inter-VLAN routing using SVIs.
- Provide centralized DHCP, DNS, Web, NTP and Syslog services.
- Use DHCP relay from user VLANs to the centralized DHCP server.
- Secure administrative access with SSH and local authentication.
- Apply ACLs, Port Security and unused-port hardening.
- Implement static edge routing and PAT at the enterprise boundary.
- Validate services and security policies with real Packet Tracer tests.
- Produce documentation suitable for a technical portfolio and GitHub repository.

1.3 Scope

Component	Quantity / Scope
Headquarters sites	1
Edge routers	1
Layer 3 core switches	1
Layer 2 access switches	1
Server nodes	3
Representative client PCs	12
Business/service VLANs	5 (10, 20, 30, 40, 50)
Parking VLAN	999
Employee population represented	Approximately 75

Scope boundary The project demonstrates a production-inspired architecture in a simulation environment. High availability, dynamic WAN routing, branch connectivity and advanced Layer 2 protection are intentionally deferred to Projects 2 and 3.

2. Business Requirements

Requirement	Design Response
Department separation	HR, Finance, IT, Guest and Server resources must remain logically separated while using shared switching infrastructure.
Secure management	Network devices must be remotely manageable using SSH rather than Telnet.
Centralized services	Addressing, naming, internal web access, time and logging should be centrally managed.
Controlled access	Security policies must restrict selected cross-VLAN flows and isolate Guest users from internal enterprise resources.
Operational visibility	Syslog and NTP support consistent event recording and troubleshooting.
Scalability	The headquarters design must provide a clean foundation for a future branch office, OSPF, VLSM and redundancy.

3. Network Design

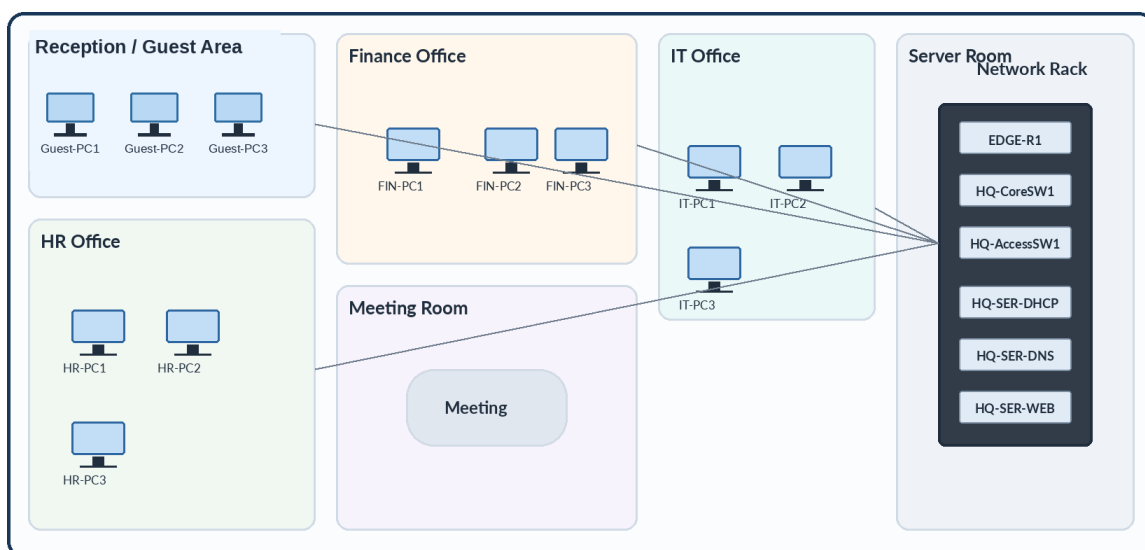
3.1 Architecture

The design follows a compact hierarchical enterprise model. Core and distribution functions are combined on a multilayer switch, while endpoint switching remains on a Layer 2 access switch. This is a common approach for small-to-medium environments where dedicated core and distribution tiers would add complexity without providing useful benefit at the current scale.

3.2 Physical Topology

The physical layout is documented as a conceptual headquarters floor plan. User areas are separated by business function, while routing, switching and server infrastructure are centralized in a server room and rack for easier administration and future handover.

Enterprise HQ - Conceptual Physical Layout



Conceptual floor-plan view for documentation. Network equipment is centralized in the Server Room / Rack.

Figure 3.1 - Enterprise Physical Topology

Conceptual physical layout used for documentation. It represents the intended placement of departments and the centralized network rack rather than a Packet Tracer workspace screenshot.

3.3 Logical Topology

The logical topology shows the actual routing and segmentation relationships: user VLANs connect through the access switch, the Layer 3 core provides the SVIs and internal routing, server resources reside in VLAN 50, and the edge router forms the external boundary for PAT and future WAN expansion.

Enterprise HQ - Logical Network Topology

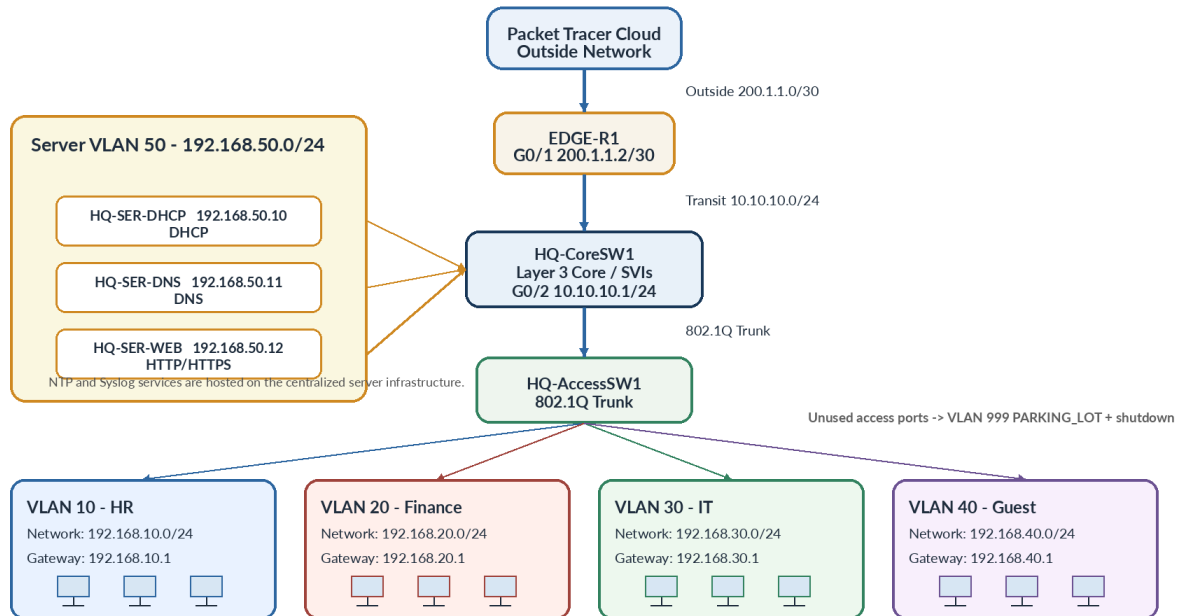


Figure 3.2 - Enterprise Logical Topology

Logical view of the implemented HQ architecture, including VLAN networks, server addressing, transit network and the simulated outside-network boundary.

3.4 Design Decisions

Decision	Engineering Rationale
Layer 3 core instead of Router-on-a-Stick	Inter-VLAN traffic is routed directly on the multilayer switch, reducing dependence on the edge router for internal traffic and providing a more scalable foundation.
Dedicated access layer	Endpoint membership, access VLANs and Port Security are concentrated on the access switch.
Edge router retained	The router provides the outside boundary, static external routing role, PAT and a future path to branch/WAN connectivity.
Centralized services	Infrastructure services are hosted in the Server VLAN rather than configured separately on every client or switching device.

3.5 Device Inventory and Naming

Device / Role	Name	Purpose
Edge Router	EDGE-R1	External boundary, static routing, NAT/PAT
Layer 3 Core	HQ-CoreSW1	SVIs, inter-VLAN routing, core switching
Layer 2 Access	HQ-AccessSW1	Endpoint access, VLAN membership, port security
DHCP Server	HQ-SER-DHCP	Centralized DHCP; 192.168.50.10
DNS Server	HQ-SER-DNS	DNS service; 192.168.50.11
Web Server	HQ-SER-WEB	Internal HTTP/HTTPS; 192.168.50.12
Client endpoints	HR / Finance / IT / Guest PCs	Representative enterprise users

3.6 Interface and Port Map

The following port map records the principal endpoint groups and inter-device links used in the implemented headquarters topology. It is intended to support troubleshooting, handover and later branch expansion.

Device	Interface / Range	Connected To	Mode / VLAN	Purpose
HQ-AccessSW1	Fa0/1-Fa0/3	HR-PC1 to HR-PC3	Access / VLAN 10	HR endpoint access
HQ-AccessSW1	Fa0/4-Fa0/6	Finance-PC1 to Finance-PC3	Access / VLAN 20	Finance endpoint access
HQ-AccessSW1	Fa0/7-Fa0/9	IT-PC1 to IT-PC3	Access / VLAN 30	IT endpoint access
HQ-AccessSW1	Fa0/10-Fa0/12	Guest-PC1 to Guest-PC3	Access / VLAN 40	Guest endpoint access
HQ-AccessSW1	Fa0/13-Fa0/24	Unused access ports	VLAN 999 / shutdown	Parking and unused-port hardening
HQ-AccessSW1	Gi0/1	HQ-CoreSW1 Gi0/1	802.1Q trunk	Carries enterprise VLAN traffic
HQ-CoreSW1	Fa0/1-Fa0/3	DHCP / DNS / Web servers	Access / VLAN 50	Server VLAN connectivity
HQ-CoreSW1	Gi0/2	EDGE-R1 Gi0/0	Layer 3 / 10.10.10.1/24	Core-to-edge transit
EDGE-R1	Gi0/0	HQ-CoreSW1 Gi0/2	NAT inside / 10.10.10.2/24	Internal enterprise boundary
EDGE-R1	Gi0/1	Packet Tracer Cloud	NAT outside / 200.1.1.2/30	Simulated outside boundary

4. IP Addressing Plan

Project 1 uses one /24 subnet per business/service VLAN. This intentionally prioritizes clarity during the first enterprise build; VLSM optimization is reserved for Project 2 so the portfolio shows a deliberate progression in addressing design.

VLAN	Department / Role	Network	Default Gateway	DHCP / Static
10	HR	192.168.10.0/24	192.168.10.1	192.168.10.100-200
20	Finance	192.168.20.0/24	192.168.20.1	192.168.20.100-200
30	IT	192.168.30.0/24	192.168.30.1	192.168.30.100-200
40	Guest	192.168.40.0/24	192.168.40.1	192.168.40.100-200
50	Servers	192.168.50.0/24	192.168.50.1	Static infrastructure addresses
999	PARKING_LOT	No user subnet	N/A	Unused access ports

4.1 Infrastructure and Transit Addresses

Purpose	Addressing
Core-to-edge transit	10.10.10.0/24
HQ-CoreSW1 routed interface	10.10.10.1/24
EDGE-R1 inside interface	10.10.10.2/24
Outside segment	200.1.1.0/30

Purpose	Addressing
EDGE-R1 outside interface	200.1.1.2/30
DHCP server	192.168.50.10
DNS server	192.168.50.11
Web server	192.168.50.12

5. VLAN Design

VLANs reduce broadcast scope, separate business functions and provide clean policy boundaries for ACL enforcement. VLAN 999 is used as a parking VLAN for unused access ports, while the active user-facing access ports are assigned to their department VLANs.

VLAN ID	Name	Purpose
10	HR	Human Resources
20	FINANCE	Finance
30	IT	Information Technology
40	GUEST	Guest users
50	SERVERS	Infrastructure services
999	PARKING_LOT	Unused access ports

```
HQ-AccessSW1#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Gig0/2
10	HR	active	Fa0/1, Fa0/2, Fa0/3
20	Finance	active	Fa0/4, Fa0/5, Fa0/6
30	IT	active	Fa0/7, Fa0/8, Fa0/9
40	GUEST	active	Fa0/10, Fa0/11, Fa0/12
50	Servers	active	
999	PARKING_LOT	active	Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
HQ-AccessSW1#
```

Figure 5.1 - VLAN Configuration

The access switch shows department access ports assigned to VLANs 10-40 and unused ports Fa0/13-Fa0/24 placed in VLAN 999.

5.1 802.1Q Trunk

The uplink between the access switch and the core is configured as an IEEE 802.1Q trunk. The active VLAN set includes VLANs 10, 20, 30, 40, 50 and 999, allowing the access layer to forward tagged traffic toward the Layer 3 core.


```

1003 cken ring default      active
1004 fddinet-default        active
1005 trnet-default          active
HQ-AccessSW1#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Gig0/1    on        802.1q         trunking    1

Port      Vlans allowed on trunk
Gig0/1    1-1005

Port      Vlans allowed and active in management domain
Gig0/1    1,10,20,30,40,50,999

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1    1,10,20,30,40,50,999

HQ-AccessSW1#

```

Figure 5.2 - Trunk Configuration

The GigabitEthernet0/1 uplink is operational as an 802.1Q trunk and forwards the active enterprise VLANs.

6. Layer 3 Switching

HQ-CoreSW1 provides a Switched Virtual Interface (SVI) for each routed VLAN and uses the ip routing feature to route traffic internally. This keeps most interdepartmental traffic within the switching fabric instead of sending it to the edge router.

VLAN	SVI	Gateway
10	Vlan10	192.168.10.1
20	Vlan20	192.168.20.1
30	Vlan30	192.168.30.1
40	Vlan40	192.168.40.1
50	Vlan50	192.168.50.1

```
HQ-CoreSW1#show ip interface brief
Interface          IP-Address      OK? Method Status      Protocol
FastEthernet0/1    unassigned      YES unset    up          up
FastEthernet0/2    unassigned      YES unset    up          up
FastEthernet0/3    unassigned      YES unset    up          up
FastEthernet0/4    unassigned      YES unset    administratively down down
FastEthernet0/5    unassigned      YES unset    administratively down down
FastEthernet0/6    unassigned      YES unset    administratively down down
FastEthernet0/7    unassigned      YES unset    administratively down down
FastEthernet0/8    unassigned      YES unset    administratively down down
FastEthernet0/9    unassigned      YES unset    administratively down down
FastEthernet0/10   unassigned      YES unset    administratively down down
FastEthernet0/11   unassigned      YES unset    administratively down down
FastEthernet0/12   unassigned      YES unset    administratively down down
FastEthernet0/13   unassigned      YES unset    administratively down down
FastEthernet0/14   unassigned      YES unset    administratively down down
FastEthernet0/15   unassigned      YES unset    administratively down down
FastEthernet0/16   unassigned      YES unset    administratively down down
FastEthernet0/17   unassigned      YES unset    administratively down down
FastEthernet0/18   unassigned      YES unset    administratively down down
FastEthernet0/19   unassigned      YES unset    administratively down down
FastEthernet0/20   unassigned      YES unset    administratively down down
FastEthernet0/21   unassigned      YES unset    administratively down down
FastEthernet0/22   unassigned      YES unset    administratively down down
FastEthernet0/23   unassigned      YES unset    administratively down down
FastEthernet0/24   unassigned      YES unset    administratively down down
GigabitEthernet0/1 unassigned      YES unset    up          up
GigabitEthernet0/2 10.10.10.1      YES manual   up          up
Vlan1              unassigned      YES unset    administratively down down
Vlan10             192.168.10.1    YES manual   up          up
Vlan20             192.168.20.1    YES manual   up          up
Vlan30             192.168.30.1    YES manual   up          up
Vlan40             192.168.40.1    YES manual   up          up
Vlan50             192.168.50.1    YES manual   up          up
HQ-CoreSW1#
```

Figure 6.1 - Layer 3 Interface Status

All production SVIs are up/up. The core-to-edge routed link uses 10.10.10.1, while unused FastEthernet interfaces on the core are administratively down.

The core routing table contains each directly connected VLAN and the transit network. A static default route points to EDGE-R1 at 10.10.10.2, so only traffic not routed internally is forwarded toward the enterprise edge.

```
HQ-CoreSW1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.10.10.2 to network 0.0.0.0

10.0.0.0/24 is subnetted, 1 subnets
C    10.10.10.0 is directly connected, GigabitEthernet0/2
C    192.168.10.0/24 is directly connected, Vlan10
C    192.168.20.0/24 is directly connected, Vlan20
C    192.168.30.0/24 is directly connected, Vlan30
C    192.168.40.0/24 is directly connected, Vlan40
C    192.168.50.0/24 is directly connected, Vlan50
S*   0.0.0.0/0 [1/0] via 10.10.10.2

HQ-CoreSW1#
```

Figure 6.2 - Routing Table

The routing table confirms locally connected VLAN networks and a default route via 10.10.10.2.

7. Network Services

Centralized services reduce manual endpoint configuration and provide consistent enterprise operations. The project implements DHCP, DNS, internal Web, NTP and Syslog services on the server infrastructure in VLAN 50.

7.1 DHCP and DHCP Relay

The DHCP server uses separate pools for HR, Finance, IT and Guest. Each pool provides the subnet mask, default gateway and DNS server address. Because the DHCP server resides in VLAN 50 while clients reside in different broadcast domains, the user SVIs use ip helper-address 192.168.50.10 to relay DHCP requests to the centralized server.

```
interface Vlan10
ip address 192.168.10.1 255.255.255.0
ip helper-address 192.168.50.10

interface Vlan40
ip address 192.168.40.1 255.255.255.0
ip helper-address 192.168.50.10
```

HQ-SER-DHCP

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP**
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: HR

Default Gateway: 192.168.10.1

DNS Server: 192.168.50.11

Start IP Address: 192 168 10 100

Subnet Mask: 255 255 255 0

Maximum Number of Users: 101

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
Guest	192.168....	192.168....	192.168....	255.255....	101	0.0.0.0	0.0.0.0
IT	192.168....	192.168....	192.168....	255.255....	101	0.0.0.0	0.0.0.0
Finance	192.168....	192.168....	192.168....	255.255....	101	0.0.0.0	0.0.0.0
HR	192.168....	192.168....	192.168....	255.255....	101	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	192.168....	255.255....	512	0.0.0.0	0.0.0.0

Figure 7.1 - DHCP Server Configuration

The HR pool is shown in detail, while the pool list confirms separate scopes for Guest, IT, Finance and HR.

7.2 DNS Service

The DNS service maps the internal hostname `www.techsolutions.local` to the internal Web server at `192.168.50.12`. This allows users to access the intranet by name instead of using a numerical IP address.

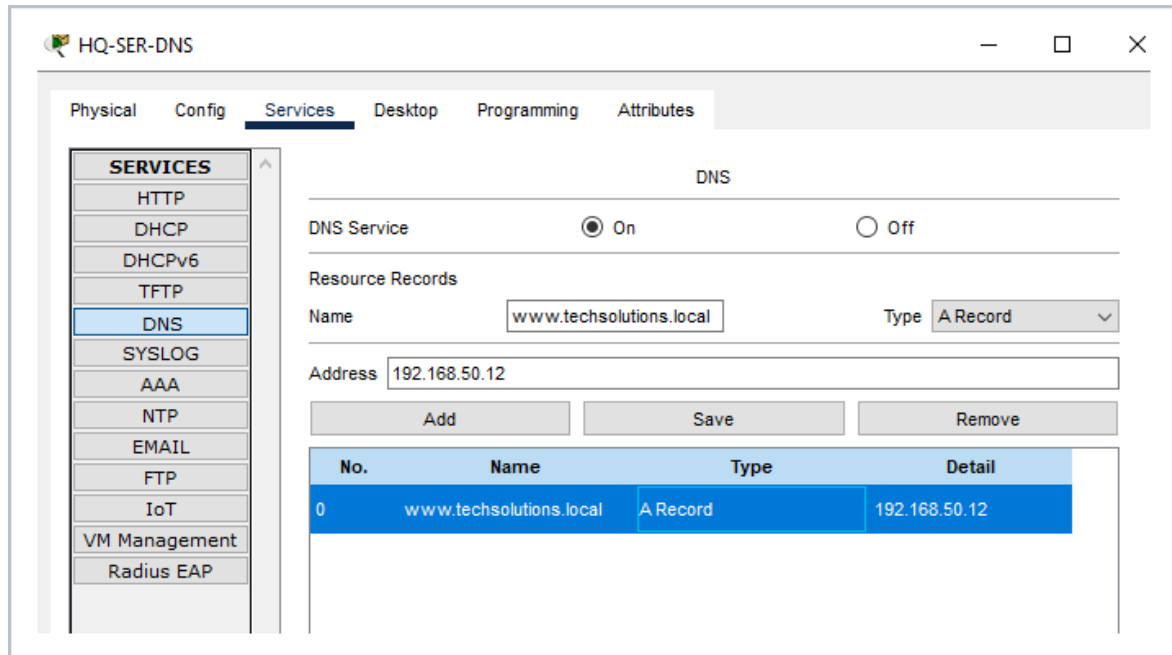


Figure 7.2 - DNS Server Configuration

The A record resolves `www.techsolutions.local` to `192.168.50.12`.

7.3 Internal Web Service

The internal Web server provides a simple TechSolutions intranet page used to validate application reachability. HTTP and HTTPS services are enabled in Packet Tracer; the validation screenshot uses HTTP through the DNS hostname.

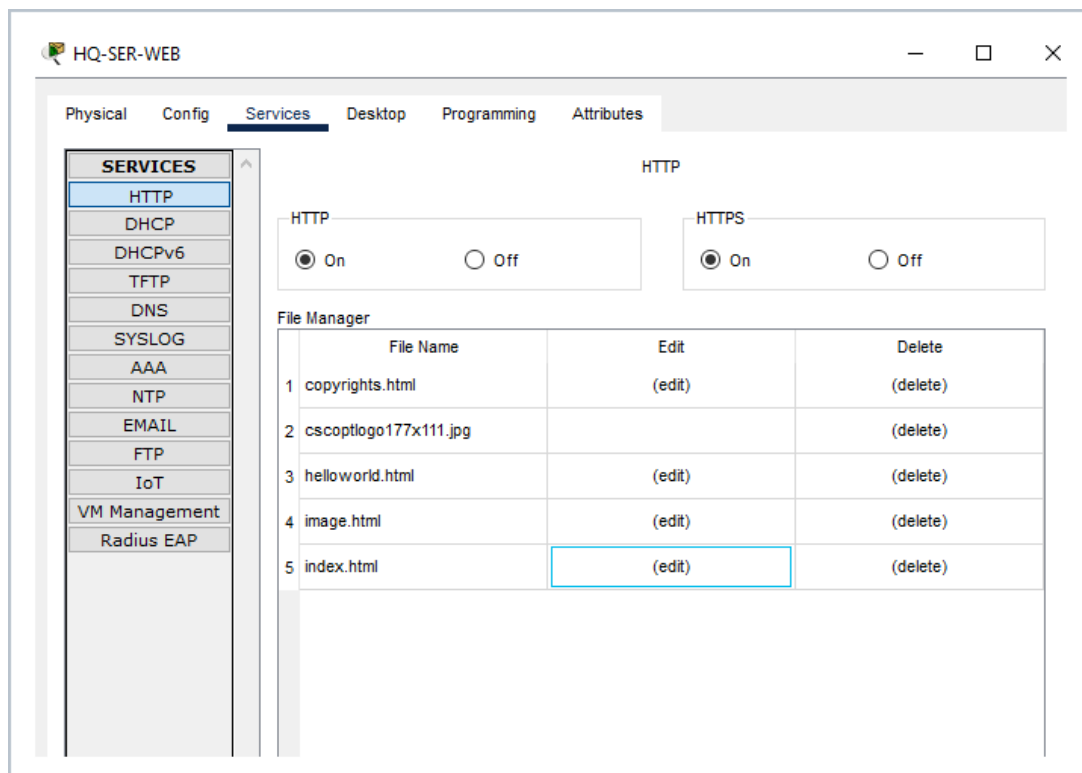


Figure 7.3 - Web Server Configuration

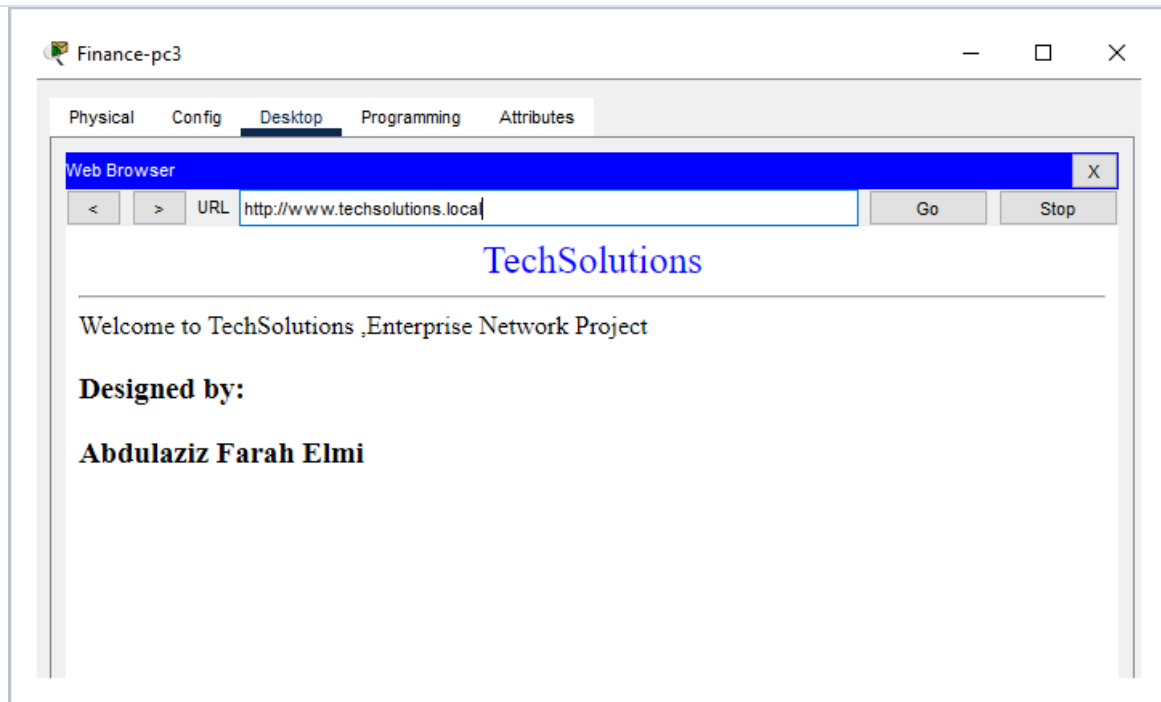


Figure 7.4 - Internal Website Access

A Finance workstation successfully opens <http://www.techsolutions.local>, validating DNS resolution, inter-VLAN routing and Web service reachability.

7.4 NTP Service

A centralized NTP service provides a common time source for network devices. Consistent time is important for troubleshooting, log correlation and later security analysis.

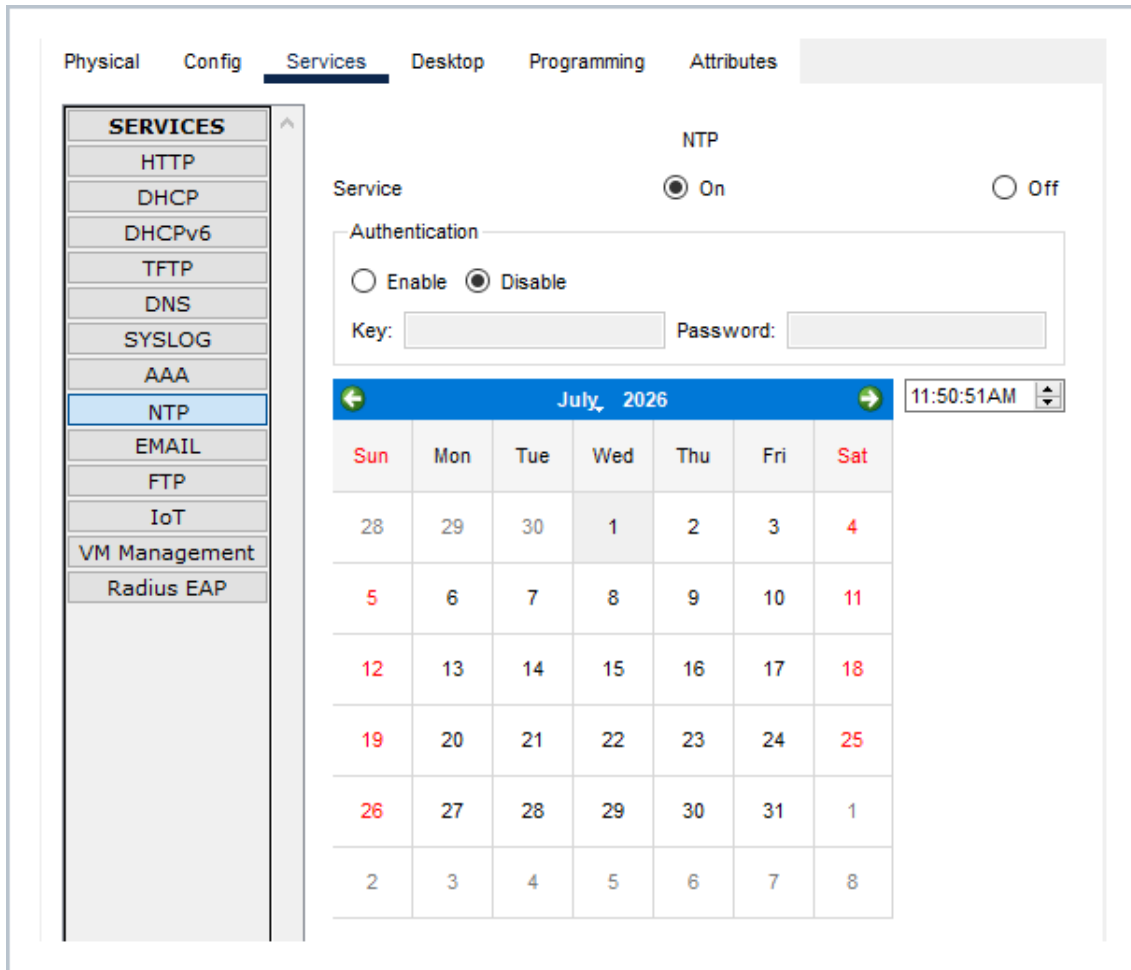


Figure 7.5 - NTP Server Configuration

The centralized NTP service is enabled in the Packet Tracer server environment.

7.5 Syslog Service

Syslog centralizes operational events instead of leaving logs only on individual devices. The captured messages show configuration events arriving at the server from the core network device.

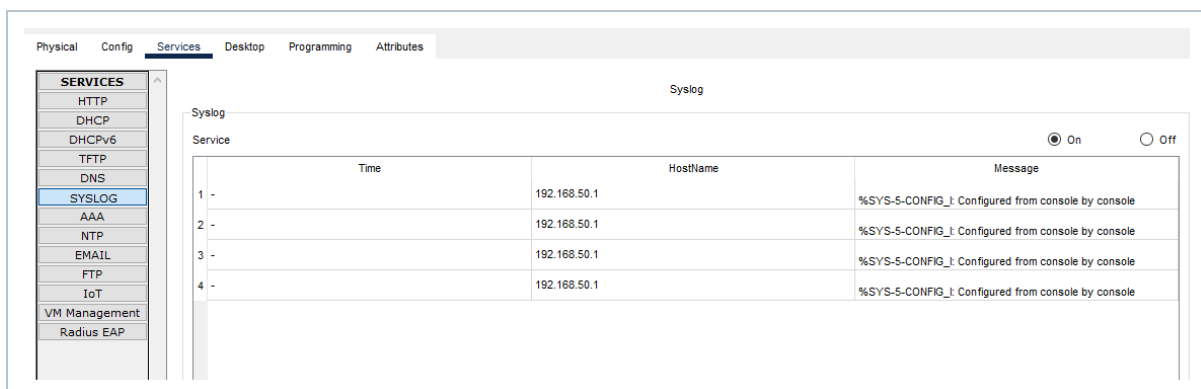


Figure 7.6 - Syslog Server Logs

Multiple configuration-event messages are received successfully, demonstrating operational centralized logging.

7.6 Service Placement Summary

Three server nodes are used in the topology. DHCP, DNS and Web have dedicated named hosts and static addresses. NTP and Syslog are documented as centralized server services because the retained screenshots validate those services but do not uniquely identify a separate dedicated host name or IP address.

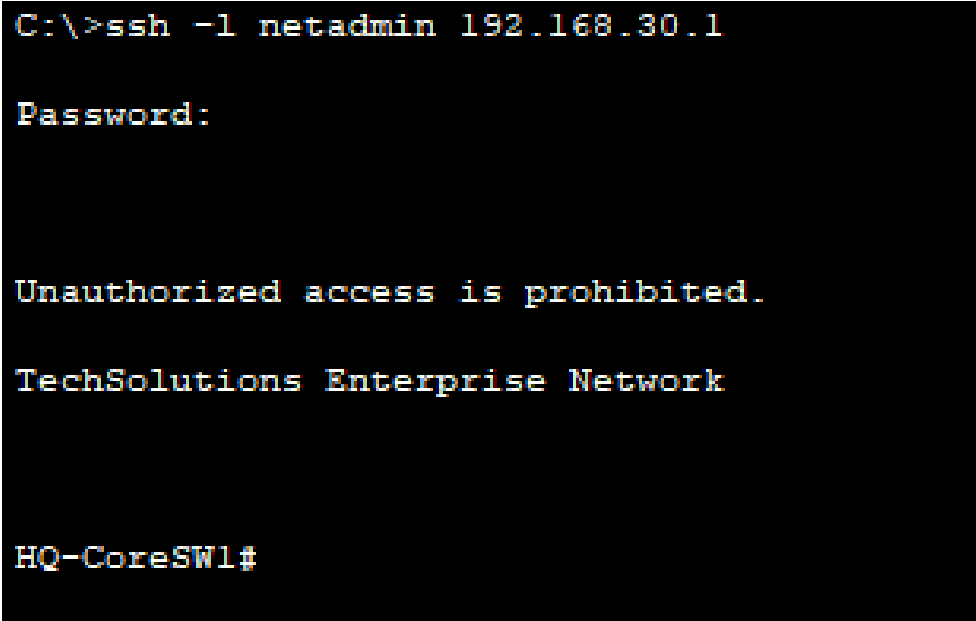
Service	Host / Placement	Address / Scope	Evidence
DHCP	HQ-SER-DHCP	192.168.50.10	Figure 7.1 + DHCP relay configuration
DNS	HQ-SER-DNS	192.168.50.11	Figure 7.2
Internal Web	HQ-SER-WEB	192.168.50.12	Figures 7.3-7.4
NTP	Centralized server infrastructure (co-hosted service)	Server VLAN service	Figure 7.5
Syslog	Centralized server infrastructure (co-hosted service)	Server VLAN service	Figure 7.6

8. Security Implementation

Security is implemented as layered controls across management access, inter-VLAN policy, access ports and unused interfaces. The goal is to reduce the attack surface while keeping legitimate business traffic operational.

8.1 Secure Device Management - SSH

Network devices use SSH for encrypted remote management with local authentication. Telnet is not used. The validation session below connects from an internal workstation to HQ-CoreSW1 and reaches privileged EXEC mode; credentials are intentionally omitted from the report.



```

C:\>ssh -l netadmin 192.168.30.1

Password:

Unauthorized access is prohibited.

TechSolutions Enterprise Network

HQ-CoreSW1#

```

Figure 8.1 - SSH Remote Login

Successful SSH access to 192.168.30.1 confirms encrypted remote administration of HQ-CoreSW1.

8.2 Access Control Lists

The implemented ACLs reflect the final working configuration rather than the earlier design draft. HR_POLICY is applied inbound on Vlan10 and blocks HR traffic destined for the IT network (192.168.30.0/24) while permitting other traffic. GUEST_POLICY is applied inbound on Vlan40 and blocks Guest traffic to HR, Finance, IT and Server networks while permitting traffic to other destinations.

Policy	Source	Restricted Destination(s)	Result
HR_POLICY	192.168.10.0/24	192.168.30.0/24 (IT)	Deny, then permit other traffic
GUEST_POLICY	192.168.40.0/24	192.168.10.0/24 (HR) 192.168.20.0/24 (Finance) 192.168.30.0/24 (IT) 192.168.50.0/24 (Servers)	Deny listed internal networks; permit other traffic

(a) Applied ACL policies

```
HQ-CoreSW1#show access-lists
Extended IP access list sl_def_acl
 0 deny tcp any any eq telnet
 0 deny tcp any any eq www
 0 deny tcp any any eq 22 (10 match(es))
 0 permit tcp any any eq 22
Extended IP access list HR_POLICY
10 deny ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255
20 permit ip any any (6 match(es))
Extended IP access list GUEST_POLICY
10 deny ip 192.168.40.0 0.0.0.255 192.168.10.0 0.0.0.255
20 deny ip 192.168.40.0 0.0.0.255 192.168.20.0 0.0.0.255
30 deny ip 192.168.40.0 0.0.0.255 192.168.30.0 0.0.0.255
40 deny ip 192.168.40.0 0.0.0.255 192.168.50.0 0.0.0.255
50 permit ip any any (6 match(es))
HQ-CoreSW1#
```

(b) HR policy verification

```
Pinging 192.168.30.1 with 32 bytes of data:
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.30.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
    C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.50.12

Pinging 192.168.50.12 with 32 bytes of data:
Reply from 192.168.50.12: bytes=32 time<1ms TTL=127
Reply from 192.168.50.12: bytes=32 time<1ms TTL=127
Reply from 192.168.50.12: bytes=32 time<1ms TTL=127
Reply from 192.168.50.12: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.50.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Configuration evidence and live endpoint behavior captured from the working Packet Tracer project.

Figure 8.2 - ACL Configuration and Verification

The applied ACL rules are shown together with live HR tests: HR-to-IT is blocked, while HR-to-Finance and HR-to-Server traffic is allowed.

8.3 Port Security

Port Security is enabled on user-facing access ports with a maximum of one secure MAC address, sticky learning and shutdown violation mode. This prevents an unrecognized endpoint from replacing the learned device without triggering a security event.


```

HQ-AccessSW1#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
      (Count)      (Count)      (Count)
-----
Fa0/1      1      1      0      Shutdown
Fa0/2      1      1      0      Shutdown
Fa0/3      1      1      0      Shutdown
Fa0/4      1      1      0      Shutdown
Fa0/5      1      1      0      Shutdown
Fa0/6      1      1      0      Shutdown
Fa0/7      1      1      0      Shutdown
Fa0/8      1      1      0      Shutdown
Fa0/9      1      1      0      Shutdown
Fa0/10     1      1      0      Shutdown
Fa0/11     1      1      0      Shutdown
Fa0/12     1      1      0      Shutdown
-----
HQ-AccessSW1#

```

Figure 8.3 - Port Security Status

Fa0/1-Fa0/12 each show one secure address and shutdown as the violation action.

A controlled test on Fa0/12 connected a different endpoint after a sticky MAC had already been learned. The port moved from Secure-up to Secure-shutdown and the violation counter incremented to 1.

```

HQ-AccessSW1#show port-security interface fa0/12
Port Security           : Enabled
Port Status             : Secure-shutdown
Violation Mode          : Shutdown
Aging Time              : 0 mins
Aging Type               : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses   : 1
Total MAC Addresses     : 1
Configured MAC Addresses : 0
Sticky MAC Addresses    : 1
Last Source Address:Vlan : 0001.6375.221C:40
Security Violation Count : 1

HQ-AccessSW1#

```

Figure 8.4 - Port Security Violation

A new source MAC triggers Secure-shutdown on Fa0/12, confirming the configured enforcement behavior.

8.4 Unused Port Hardening and Parking VLAN

Unused access ports are assigned to VLAN 999 (PARKING_LOT) and administratively disabled. Separating these ports from production VLANs and shutting them down reduces the risk of unauthorized physical connections. Figure 5.1 provides the VLAN-assignment evidence for Fa0/13-Fa0/24.

8.5 Security Summary

Control	Status	Evidence
SSH-only remote management	Implemented	Figure 8.1
Applied ACL policies	Implemented and validated	Figure 8.2
Port Security + sticky MAC	Implemented	Figure 8.3

Control	Status	Evidence
Shutdown violation behavior	Validated	Figure 8.4
Parking VLAN 999	Implemented	Figure 5.1
Unused-port shutdown	Implemented - configuration verified	Parking VLAN + configuration state; dedicated interface-status capture recommended

9. External Connectivity and NAT/PAT

9.1 Edge and Transit Design

EDGE-R1 connects the internal enterprise to the simulated outside segment. The internal side uses 10.10.10.2/24 toward HQ-CoreSW1, while the outside interface uses 200.1.1.2/30 toward the Packet Tracer Cloud. The core sends unknown destinations to 10.10.10.2, and EDGE-R1 has a summarized static route back to 192.168.0.0/16 via 10.10.10.1.

9.2 PAT Configuration

PAT is configured with GigabitEthernet0/0 as the inside interface and GigabitEthernet0/1 as the outside interface. Standard ACL 1 identifies the internal 192.168.0.0/16 address space, and the overload rule translates eligible internal traffic to the G0/1 outside address.

```
access-list 1 permit 192.168.0.0 0.0.255.255
ip nat inside source list 1 interface GigabitEthernet0/1 overload
```

(a) NAT inside/outside and PAT configuration

```
EDGE-R1#show running-config | include ip nat
ip nat inside
ip nat outside
ip nat inside source list 1 interface GigabitEthernet0/1 overload
EDGE-R1#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  10.10.10.2      YES manual up          up
GigabitEthernet0/1  200.1.1.2       YES manual up          up
GigabitEthernet0/2  unassigned      YES unset  administratively down down
Vlan1          unassigned      YES unset  administratively down down
EDGE-R1#
```

(b) NAT/PAT runtime statistics

```
EDGE-R1#show ip nat statistics
Total translations: 0 (0 static, 0 dynamic, 0 extended)
Outside Interfaces: GigabitEthernet0/1
Inside Interfaces: GigabitEthernet0/0
Hits: 0 Misses: 21
Expired translations: 16
Dynamic mappings:
EDGE-R1#
```

Figure 9.1 - NAT/PAT Configuration and Statistics

The router shows the inside/outside roles and overload rule. Runtime statistics report one dynamic mapping and sixteen expired translations, indicating that PAT translation entries were previously created and subsequently aged out; no active entry existed at the instant of capture.

Packet Tracer limitation The Cloud object used in this project represents the external boundary but is not a real Internet host that can reply to arbitrary public pings. For that reason, the final report validates PAT through configuration, interface roles and translation statistics rather than claiming live Internet reachability to a public endpoint.

10. Validation and Testing

Validation combines service evidence, CLI verification and endpoint testing. The objective is to prove that the implemented configuration produces the expected behavior rather than merely existing in the running configuration.

Test	Expected Behavior	Result
DHCP address assignment	Client receives correct subnet, gateway and DNS	PASS
DNS name resolution	www.techsolutions.local resolves to 192.168.50.12	PASS
Internal Web access	Client opens TechSolutions intranet	PASS
Inter-VLAN routing	Authorized VLANs communicate through HQ-CoreSW1	PASS
SSH management	Authorized local user reaches HQ-CoreSW1 over SSH	PASS
HR ACL enforcement	HR blocked from IT but permitted to Finance/Servers	PASS
Guest isolation policy	GUEST_POLICY applied inbound on Vlan40; internal destinations explicitly denied	CONFIG VERIFIED
Port Security	One sticky MAC per user-facing port	PASS
Port Security violation	Unauthorized MAC causes Secure-shutdown	PASS
Syslog	Centralized server receives device events	PASS
NTP service	Centralized NTP service enabled in the server environment	CONFIG VERIFIED
NAT/PAT	Inside/outside roles and overload rule present; prior translation entries aged out	CONFIG + STATISTICS VERIFIED

10.1 End-to-End Connectivity

The final connectivity test uses an IT client with address 192.168.30.102/24 and default gateway 192.168.30.1. The client reaches the Finance SVI at 192.168.20.1 and the Web server at 192.168.50.12 with 0% packet loss, demonstrating correct DHCP addressing and routed reachability across authorized VLANs.

(a) IT client addressing

```
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::250:FFF:FE91:A6D2
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.30.102
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                   192.168.30.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0
```

(b) Inter-VLAN end-to-end connectivity

```
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.50.12

Pinging 192.168.50.12 with 32 bytes of data:

Reply from 192.168.50.12: bytes=32 time<1ms TTL=127
Reply from 192.168.50.12: bytes=32 time<1ms TTL=127
Reply from 192.168.50.12: bytes=32 time<1ms TTL=127
Reply from 192.168.50.12: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.50.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 10.1 - End-to-End Connectivity Test

10.2 Validation Boundaries and Assumptions

The report distinguishes directly validated behavior from configuration-only verification so that the portfolio does not overstate what Packet Tracer or the retained evidence proves.

Boundary / Assumption	How It Is Treated in This Report
Simulation scale	Twelve representative PCs model an organization of approximately 75 employees; the addressing design leaves room for additional hosts.
Outside connectivity	The Packet Tracer Cloud represents the external boundary, not a live Internet host. Public Internet reachability is therefore not claimed.
NAT/PAT evidence	Configuration, interface roles and translation statistics are verified. Expired translations indicate prior PAT entries that aged out.
NTP evidence	The centralized NTP service is enabled; device-by-device synchronization is not independently demonstrated in the retained screenshots.
Guest isolation evidence	GUEST_POLICY is applied inbound on Vlan40 and its deny rules are configuration-verified. A dedicated endpoint denial test can be added during the final lab-hardening pass.
Availability	Project 1 intentionally uses single edge/core/access devices; redundancy is deferred to Project 3.
Addressing efficiency	Project 1 uses /24 networks for clarity. VLSM is intentionally reserved for Project 2.

11. Troubleshooting

Issue 1 - Configuration Loss

Problem: Packet Tracer was closed before the active configuration had been preserved.

Cause: Running configuration had not been copied to startup configuration.

Resolution: Devices were rebuilt and regular copy running-config startup-config saves were incorporated into the workflow.

Lesson learned Configuration backups and save checkpoints are essential, even in simulation.

Issue 2 - Port Security / Err-disabled

Problem: A secured access port disabled when a different endpoint was connected.

Cause: Sticky MAC had already learned the original endpoint and violation mode was shutdown.

Resolution: The behavior was verified as intentional; the port was restored only after the test and violation condition were addressed.

Lesson learned Security controls must be understood before treating their enforcement as a fault.

Issue 3 - SSH Prerequisites

Problem: SSH initially refused remote access during the project build.

Cause: Required hostname/domain/RSA/local-user/VTY prerequisites were incomplete.

Resolution: The required SSH prerequisites were configured and remote access was validated from a client PC.

Lesson learned SSH is a dependency chain, not a single command.

Issue 4 - DHCP Verification

Problem: Clients initially failed to obtain the expected addressing.

Cause: DHCP pool gateway values did not match the VLAN gateway design.

Resolution: Scope values were corrected and client leases were renewed.

Lesson learned Centralized services must be validated against the IP plan, not configured in isolation.

12. Lessons Learned

Area	Practical Outcome
Network design	Translated business requirements into a structured topology, addressing plan and staged portfolio architecture.
Switching and routing	Built VLANs, trunks, SVIs, Layer 3 switching and static edge routing, then verified them using operational commands.
Network services	Integrated DHCP, DHCP relay, DNS, Web, NTP and Syslog into one enterprise scenario.
Security	Applied management hardening, ACLs, Port Security, sticky MAC learning and unused-port controls.
Troubleshooting	Used outputs, expected-vs-actual behavior and controlled testing to isolate configuration issues.
Documentation	Converted the working lab into a technical report with implementation evidence and stated limitations.

13. Future Improvements

The portfolio is intentionally staged. Project 1 establishes a functional HQ; Project 2 expands the company to a branch office; Project 3 introduces resiliency and advanced protection.

Next Stage	Planned Technologies	Purpose
Project 2 - HQ + Branch	VLSM, WAN link, OSPF, expanded DHCP relay, inter-site ACLs	Introduce efficient addressing and dynamic multi-site routing.
Project 3 - Enterprise Expansion	RSTP, EtherChannel, gateway redundancy/HSRP if supported, DHCP Snooping, stronger Layer 2 security, optional site-to-site VPN	Introduce resiliency, availability and deeper enterprise security.

Design progression VLSM and redundancy are not omissions from Project 1; they are intentionally deferred so each portfolio project demonstrates a clear increase in engineering complexity.

14. Conclusion

Project 1 successfully demonstrates the end-to-end design, configuration and validation of a segmented enterprise headquarters network in Cisco Packet Tracer. The completed architecture includes departmental VLANs, Layer 3 switching, centralized services, DHCP relay, SSH management, ACL enforcement, Port Security, centralized logging, static edge routing and PAT configuration.

The project also documents its simulation limitations rather than overstating them. Internal service delivery and security behavior were validated directly, while the outside Packet Tracer Cloud was treated as a simulated boundary rather than a live Internet endpoint. This distinction improves the technical credibility of the portfolio.

Most importantly, the project provides a stable baseline for the next stages of the enterprise portfolio: branch connectivity and dynamic routing in Project 2, followed by redundancy and advanced security in Project 3.

Appendix A - Verification Commands

A.1 EDGE-R1

```
show ip interface brief
show running-config
show ip route
show access-lists
show ip nat translations
show ip nat statistics
```

A.2 HQ-CoreSW1

```
show vlan brief
show interfaces trunk
show ip interface brief
show ip route
show running-config
show access-lists
show mac address-table
```

A.3 HQ-AccessSW1

```
show vlan brief
show interfaces trunk
show interfaces status
show port-security
show port-security interface fa0/12
show mac address-table
```

Appendix B - Evidence Checklist

Figure	Evidence	Status
3.1	Conceptual physical topology	Included
3.2	Logical topology	Included
5.1	VLAN configuration	Included
5.2	802.1Q trunk	Included
6.1	Layer 3 interface status	Included
6.2	Routing table	Included
7.1	DHCP configuration	Included
7.2	DNS record	Included
7.3	Web server configuration	Included
7.4	Internal website access	Included
7.5	NTP service	Included - service enabled
7.6	Syslog messages	Included
8.1	SSH remote login	Included
8.2	ACL configuration and verification	Included
8.3	Port Security status	Included
8.4	Port Security violation	Included

Figure	Evidence	Status
9.1	NAT/PAT configuration and statistics	Included - config + statistics
10.1	End-to-end connectivity	Included

Appendix C - GitHub Repository Structure

```

Enterprise-Network-Portfolio/
├── Project-1-Enterprise-HQ/
│   ├── Enterprise-HQ-Network.pkt
│   ├── Enterprise-HQ-Network-Report.pdf
│   ├── Configurations/
│   │   ├── EDGE-R1.txt
│   │   ├── HQ-CoreSW1.txt
│   │   └── HQ-AccessSW1.txt
│   └── Images/
│       ├── 01-Physical-Topology.png
│       ├── 02-Logical-Topology.png
│       ├── 03-VLAN-Configuration.png
│       ├── 04-Trunk-Configuration.png
│       ├── 05-Layer3-Interfaces.png
│       ├── 06-Routing-Table.png
│       ├── 07-DHCP-Configuration.png
│       ├── 08-DNS-Configuration.png
│       ├── 09-Web-Server.png
│       ├── 10-Website-Access.png
│       ├── 11-NTP-Service.png
│       ├── 12-Syslog-Messages.png
│       ├── 13-SSH-Remote-Login.png
│       ├── 14-ACL-Verification.png
│       ├── 15-Port-Security-Status.png
│       ├── 16-Port-Security-Violation.png
│       ├── 17-NAT-PAT-Statistics.png
│       └── 18-End-to-End-Connectivity.png
├── Project-2-HQ-Branch/
├── Project-3-Enterprise-Expansion/
└── README.md

```

Appendix D - Technical References

Reference	Relevance to Project 1
Cisco Documentation	IOS and Catalyst configuration guidance for VLANs, trunks, inter-VLAN routing, SSH, ACLs, Port Security and NAT/PAT (cisco.com).
IEEE 802.1Q	IEEE Standard for Local and Metropolitan Area Networks - Bridges and Bridged Networks; VLAN tagging and trunking concepts.
IETF RFC 1918	Address Allocation for Private Internets; private IPv4 address space used throughout the enterprise LAN.
IETF RFC 2131	Dynamic Host Configuration Protocol (DHCP); foundation for automated client address assignment.
IETF RFC 1034 / RFC 1035	Domain Names - Concepts and Facilities / Implementation and Specification; DNS fundamentals.
IETF RFC 5905	Network Time Protocol Version 4; background for centralized time synchronization.
IETF RFC 5424	The Syslog Protocol; background for centralized event logging.
Cisco Packet Tracer	Simulation platform used to implement and validate the lab scenario; behavior is documented with stated simulation limitations.

Acknowledgment

This project was developed as part of a structured self-learning journey in Cisco networking and enterprise infrastructure. The objective was not only to execute commands, but to understand the engineering decisions behind the design, troubleshoot real configuration behavior, validate each major feature and produce documentation that reflects practical skills relevant to network engineering, NOC, infrastructure and network-security roles.